tmfem: a validated two-dimensional thermo-mechanical finite-element solver

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Parts that a particle beam hits on purpose, such as targets and collimators, are designed in two steps: a Monte Carlo code such as FLUKA [1] computes where the beam deposits its energy, then a finite-element code computes the resulting temperature and thermal stress. tmfem rebuilds the second step from scratch in C++17, without any finite-element library, and checks every capability with an automated test against an analytical solution or a reference value.

1 Methods

tmfem solves transient heat conduction and diffusion, $\rho c\,\dot{T} = \nabla \cdot (k \nabla T) + q$, in 1D and 2D with the θ-method (backward Euler or Crank–Nicolson). It solves 2D linear elasticity in plane stress and plane strain with three-node (T3) and six-node (T6) triangles, using the stiffness matrix

$$\mathbf{K} = \sum_{e} \int_{\Omega_e} \mathbf{B}^T \mathbf{D}\, \mathbf{B}\, \mathrm{d}A. \tag{1}$$

Plasticity follows the von Mises model with isotropic hardening, integrated by the radial return algorithm with a consistent tangent [2], and solved with Newton's method. The temperature field enters the mechanical problem as a thermal strain $\alpha(T - T_{\mathrm{ref}})$. Meshes are generated with Gmsh [3], and linear systems are solved with Eigen.

2 Validation

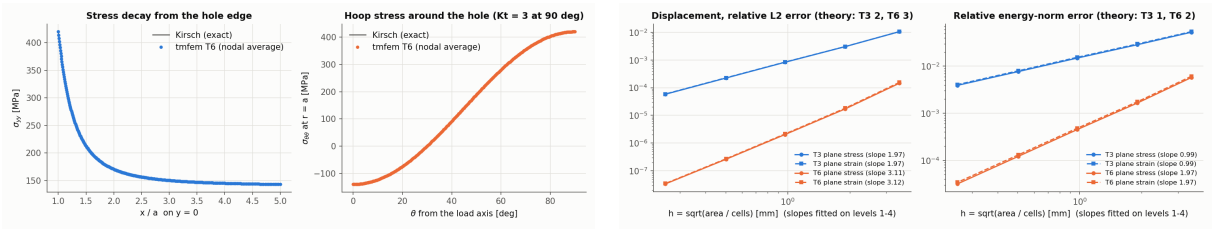
The benchmarks come from lab exercises in diffusion, heat transfer and mechanics of materials (M. R. Kabiri, ENSAM Meknès), from textbook solutions [4, 5], and from an Abaqus simulation. Every reference value was recomputed independently before being used as a test target. Table 1 summarises the main results.

Benchmark	Quantity	Reference	tmfem
Carburizing (diffusion)	time to reach 0.8 wt% at 0.5 mm	25 370 s	25 374 s
Steel plate (1D heat)	difference to a lumped model	–	below 0.033 K
Plate with a hole (Kirsch)	stress concentration factor	3	2.9997
Plate with a hole (Kirsch)	largest stress error, $r/a = 2,4$	0	0.0044 MPa
Plate with a hole, T6	convergence order (L^2 / energy)	3 / 2	3.11 / 1.97
Tensile test (vs Abaqus)	strains ε_{yy}, ε_{xx} at 457 MPa	0.03158, −0.01535	0.031581, −0.015350
Plasticity solver	Newton convergence rate	2	2.12
Heated thin disk	thermal stress error	0	below 0.04%
Heated thin disk, T6	convergence order (L^2 / energy)	3 / 2	3.06 / 2.04

Table 1: Main validation results. The full list is in the repository README.

The stress concentration at the hole matches the exact value of 3 to 0.01%, and the stresses around it match Kirsch's solution (Figure 1). The observed convergence orders match theory for

both element types. The plasticity solver reproduces the Abaqus strains of the tensile test, and the quadratic convergence of Newton's method confirms that the tangent is consistent. When the plate with a hole is loaded to 140 MPa, the material yields at the edge of the hole and a plastic zone of 0.17 times the hole radius forms. The edge strain is 11% below the estimate of Neuber's rule [6], which is known to overestimate notch strains.



(a) Stresses around the hole

(b) Convergence with mesh size

Figure 1: Plate with a hole: finite-element results against Kirsch's exact solution.

3 Demonstration: a beam-heated block

As a demonstration, a Gaussian beam deposits 1 kJ per metre in 10 μs into a long block of copper or graphite, and tmfem computes the temperature and the thermal stress (Figure 2). For a 1 mm beam, graphite heats up 2.65 times more than copper (123 K against 46.2 K) but carries 20 times less stress (3.7 against 74.3 MPa). Copper exceeds its yield stress for beams of 1 mm or smaller, while graphite stays far below its strength. This illustrates why graphite is used to absorb beams.

4 Conclusion

tmfem implements the main methods behind a thermal-stress analysis of a beam-heated component, and each of them is confirmed by an automated test against an independent answer: analytical solutions, textbook results, and an Abaqus reference.

Acknowledgement. Benchmark cases adapted from lab materials by M. R. Kabiri, ENSAM Meknès, with permission.

References

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[2] J. C. Simo and T. J. R. Hughes. *Computational Inelasticity*, volume 7 of *Interdisciplinary Applied Mathematics*. Springer, New York, 1998. doi: 10.1007/b98904.

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[4] J. Crank. *The Mathematics of Diffusion*. Clarendon Press, Oxford, 2nd edition, 1975.

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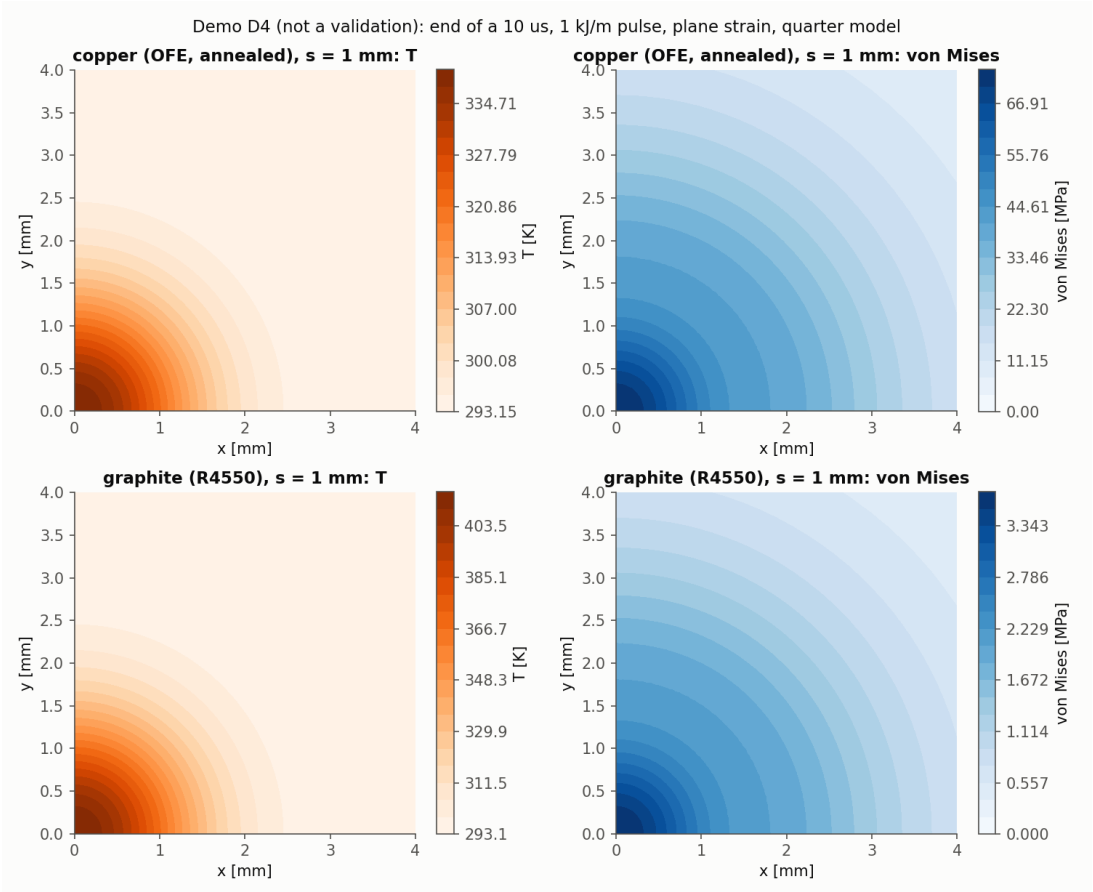


Figure 2: Temperature and von Mises stress at the end of the beam pulse.

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